HE2003 Econometrics I
Tutorial Question Set No.10
To be discussed on Nov 4 (Monday), 2024

1. Explain the following terms; Use specific example, formula, or graph if necessary.
   (a) Explain the similarity and disimilarity between t-test and homoskedasticity-only F-test.
   (b) Explain the the overall regression F-test in the homoskedastic case.
   (c) Given an empirical example of linear regression model with two dummy variables and their interaction term (the model also includes a constant term). Explain the meaning of all the parameters in the model.

2. The following regression studies the tradeoff between time spent sleeping and working after controlling for education and age:

$$\widehat{sleep} = \underset{(112.28)}{3638.25} - \underset{(0.017)}{0.148}\, totalwork - \underset{(5.88)}{11.13}\, educ + \underset{(1.45)}{2.20}\, age, \; n = 706, \; R^2 = 0.113, \quad (1)$$

where *sleep* and *totalwork* are measured in minutes per week, while *educ* and *age* are measured in years. Dropping *educ* and *age* from the equation gives

$$\widehat{sleep} = \underset{(38.91)}{3586.38} - \underset{(0.017)}{0.151}\, totalwork, \; n = 706, \; R^2 = 0.103, \quad (2)$$

Suppose we want to test the joint null hypothesis that the coefficients of *educ* and *age* are zero. Compute the value of the homoskedasticity-only F-test using the R^2 from the two regressions. To what kind of distribution and what critical value should we compare the value of the F-test? Can we reject the null hypothesis at the 5% level? Explain.

3. This question refers to the table below, computed using data on employees in a developing country. The dataset consists of information over 10,973 full-time, full-year workers. The highest educational achievement for each worker is a high school diploma. The workers' ages range from 25 to 40 years. The dataset also contains information on the region of the country where the person lives, gender, and age.

   Let AWE = logarithm of average weekly wage (in the country's currency), $High\ School$ = binary variable (1 if high school, 0 if less), $Male$ = binary variable (1 if male, 0 if female), Age = (in years), $North$ = binary variable (1 if Region = North, 0 otherwise), $East$ = binary variable (1 if Region = East, 0 otherwise), $South$ = binary variable (1 if Region = South, 0 otherwise), and $West$ = binary variable (1 if Region = West, 0 otherwise).

(a) Using the regression results in column (1):

i. Is the high school earning difference estimated from this regression statistically significant at the 5% level? Explain. Construct a 95% confidence interval of the difference.

ii. Is the male-female earning difference estimated from this regression statistically significant at the 5% level? Explain. Construct a 95% confidence interval of the difference.

Results of Regressions of Average Weekly Earnings on Gender and Education Binary Variables and Other Characteristics Using 2007 Data from a Developing Country Survey

Dependent variable: log average weekly earnings (*AWE*).

Regressor	(1)	(2)	(3)
High school graduate (X_1)	0.352 (0.021)	0.373 (0.021)	0.371 (0.021)
Male (X_2)	0.458 (0.021)	0.457 (0.020)	0.451 (0.020)
Age (X_3)		0.011 (0.001)	0.011 (0.001)
North (X_4)			0.175 (0.037)
South (X_5)			0.103 (0.033)
East (X_7)			−0.102 (0.043)
Intercept	12.84 (0.018)	12.471 (0.049)	12.390 (0.057)
Summary Statistics and Joint Tests			
F-statistic for regional effects = 0			21.87
SER	1.026	1.023	1.020
R^2	0.0710	0.0761	0.0814
n	10973	10973	10973

(b) Using the regression results in column (2):

i. Is age an important determinant of earnings? Use an appropriate statistical test and confidence interval to explain your answer.

ii. Suppose Alvo is a 30-year-old male high-school graduate, and Kal is a 40-year-old male high-school graduate. Construct a 95% confidence interval for the predicted difference between their earnings.

[Hint: Consider changing X_1 by a given amount, ΔX_1. The expected change in Y associated with this change in X_1 is $\beta_1 \Delta X_1$. Then, given the estimate $\hat{\beta}_1$, the

95% confidence interval for the effect of changing X_1 by ΔX_1 can be expressed as:

$$\begin{aligned} & 95\% \text{ CI for } \beta_1 \Delta X_1 \\ = \; & [(\hat{\beta}_1 - 1.96 \cdot SE(\hat{\beta}_1))\Delta X_1, \; (\hat{\beta}_1 + 1.96 \cdot SE(\hat{\beta}_1))\Delta X_1]. \;\;] \end{aligned}$$

(c) Using the regression results in column (3): Are there any important regional difference? Use an appropriate statistical test to explain your answer. In particular, write down the details of the null and alternative hypotheses (with the appropriate regression models), the formula of the test statistic, the critical value, and your conclusion.

4. (a) An economist in US is interested in the impact of Wisconsin's 2011 "Act 10" legislation on wages (which reduced the power of labour unions). She computes the following statistics for average hourly wage rates in Wisconsin and the neighbouring state of Minnesota for the decades before and after Act 10 was enacted (see the table below). Then, she estimates the effect of the "Act 10" legislation on wages by using the following formula:

$$(16.72 - 15.23) - (18.10 - 16.42) = -0.19. \tag{3}$$

Does this estimation formula make sense for you? What would be the key assumption involved in the estimation? Explain.

	Years	Average Wage
Wisconsin	2001-2010	15.23
Wisconsin	2010-2020	16.72
Minnesota	2001-2010	16.42
Minnesota	2010-2020	18.10

(b) Give an empirical example of using the "difference-in-differences" (DID) to study the effect of certain policy change or intervention. Show the diagram of the DID design in your example.

5. Empirical Exercise: Use the dataset **growth.dta** uploaded in NTULearn, which contains data on average economic growth rate for 65 countries, along with variables that are potentially related to growth. A detailed description of the variables is given in the table below.

(a) Construct a scatterplot of average annual growth rate ($growth$) and average trade share ($tradeshare$). For example, for STATA we can use: **scatter growth tradeshare**

Suppose that we want to run a regression of $growth$ on $tradeshare$. According to the scatterplot, one country, Malta, has a tradeshare much larger than the other countries. Explain how Malta may affect the regression result.

(b) Run a multiple regression of *growth* on the following variables: *tradeshare*, *yearsschool*, *rev_coups*, *assasinations*, and *rgdp*60. What is the estimated value of the coefficient on *yearsschool*? Is the coefficient significant at the 5% level? Explain using t-test, p-value and confidence interval.

(c) Why is *oil* omitted from the regression? What would happen if it were included?

(d) Test the joint hypothesis that none of the political/criminal factors has effect on economic growth rate. What is your conclusion?

Variable Definitions

Variable	Definition
Country_name	Name of country
growth	Average annual percentage growth of real Gross Domestic Product (GDP)* from 1960 to 1995.
rgdp60	The value of GDP* per capita in 1960, converted to 1960 US dollars
tradehare	The average share of trade in the economy from 1960 to 1995, measured as the sum of exports plus imports, divided by GDP; that is, the average value of $(X + M)/GDP$ from 1960 to 1995, where X = exports and M = imports (both X and M are positive).
yearsshcool	Average number of years of schooling of adult residents in that country in 1960
rev_coups	Average annual number of revolutions, insurrections (successful or not) and coup d'etats in that country from 1960 to 1995
assasinations	Average annual number of political assassinations in that country from 1960 to 1995 (per million population)
oil	= 1 if oil accounted for at least half of exports in 1960 = 0 otherwise